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Employee Mental Health in Offshore Operations: A Proactive Approach to Monitoring and Preventing Risks

S. R. Fernanda and R. F. F. Josenildo, CEPEM Business Psychology

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Abstract

This paper aims to present the rationale, structure, and results of the pre-flight check-in program, implemented in Brazilian offshore companies starting in 2021 as a preventive mental health tool. The scope includes an analysis of the program's effectiveness in the early identification of psychological distress, the continuous management of psychosocial risks, and the reduction of critical events and incidents related to the mental state of workers. The pre-flight check-in program consists of an online psychological screening questionnaire administered in the days leading up to embarkation. Cases considered to have moderate or alert vulnerabilities are referred for a follow-up with a psychologist for a more in-depth case analysis and support.

Confirmed severe vulnerabilities trigger an intervention protocol jointly with the company's medical department. The collected data are systematized into managerial dashboards that allow for analysis by function, department, and vessel, enabling specific strategic interventions for the most vulnerable contexts. Since its implementation, the program has demonstrated a positive impact on both prevention and rapid response to critical cases.

The initiative has contributed to reducing the embarkation of professionals with high emotional distress and to building a culture of psychological safety and preventive care. The pre-flight check-in proposal represents a paradigm shift in how mental health is addressed in the oil and gas industry. By anticipating the effects of psychosocial risks, the program directly benefits companies and employees by fostering safer and more resilient environments. It is an innovation with high potential for replicability and adaptability for offshore, onshore, and high-complexity operations.

Keywords: Fatigue Monitoring, Psychosocial risk, Risk reduction, Operational Safety, Employee wellbeing

Introduction

Workforce mental health has become a priority in the global corporate landscape. The COVID-19 pandemic amplified discussions around psychological illness, highlighting the fragility of traditional psychological care strategies and the urgent need for structured, continuous, and preventive actions. In the Oil and Gas sector, especially in offshore operations, the challenge is intensified by geographic isolation, a confined routine, operational risks, family separation, and the high technical and emotional demands placed on workers.

These factors create fertile ground for the emergence and intensification of psychological distress, which is often invisible within traditional corporate practices. In this context, since 2021, a pioneering mental health monitoring and management program called Pre-Flight Check-In was implemented in Brazil, proposing a proactive and continuous approach, anchored in emotional assessments conducted before the embarkation of offshore workers.

The program stands out by breaking with the traditional reactive logic—centered on waiting for spontaneous requests for help—and establishing an active channel for listening, screening, care, and decision-making. This paper presents the rationale, structure, methodology, and key results of the program, based on more than one hundred and fifty thousand (150.000) check-ins conducted over four years. Furthermore, the organizational implications of the strategy, its methodological innovation, its impacts on operational safety, and its potential for replicability in other critical operations are discussed.

Theoretical Background

The management of mental health in the workplace has undergone profound transformations in recent decades, driven by social changes, scientific advancements, and new international standards. The World Health Organization (WHO) estimates that more than 15% of working adults experience some form of mental disorder during their working life, with anxiety and depression being the most prevalent disorders (WHO, 2022).

In the offshore industry, these numbers take on a more serious dimension. A survey by the International SOS Foundation, widely publicized by the IADC (International Association of Drilling Contractors), revealed that up to 40% of workers in a remote rotational regime (onshore and offshore) reported having suicidal thoughts at some point while at work, and almost a third met the criteria for clinical depression while on rotation.

This scenario highlights the urgent need for proactive interventions such as the Pre-Flight Check-In program. Brazilian legislation has advanced in recognizing and regulating psychosocial risks in the workplace. The new version of NR-01, effective from August 2024, requires organizations to conduct continuous assessments of these risks, explicitly including them in their Risk Management Programs (PGRs).

Concurrently, international standards like ISO 45003 reinforce the need to incorporate mental health into organizational safety and wellness policies, focusing on prevention and systematic care.

Laws, standards, and labor regulations, at both global and national levels, have progressively reflected the importance of considering psychosocial risk factors in the organizational dimension. This movement was largely driven by a growing body of evidence linking psychological distress to decreased productivity, increased accidents, and higher turnover rates. The ISO 45001 (2018) standard on occupational health and safety management systems initiated this movement by including active worker participation and attention to mental health in its prevention framework. Its emphasis on identifying hazards and assessing risks is now understood to encompass psychosocial factors, such as high workload, lack of control, and poor social support.

This marked a critical shift from a purely physical-safety focus to a more holistic view of worker well-being. In 2021, the release of ISO 45003, the first global standard specifically addressing psychological health and safety at work, provided a comprehensive framework for organizations to implement the principles of ISO 45001 in this domain. ISO 45003 offers clear guidelines on how to identify and manage psychosocial risks and hazards. It specifies the need for a collaborative approach involving workers and their representatives in the design and implementation of psychosocial risk management systems.

The standard also underscores the importance of a positive organizational culture that promotes psychological safety and supports employees' mental well-being. The recognition of burnout by the WHO in 2022 as an occupational phenomenon further solidified this agenda worldwide, providing a common international language for discussing and addressing the issue and prompting employers to take a more preventative stance.

The International Labour Organization (ILO) has also been a key player in shaping this discourse. Its conventions and recommendations, such as the Occupational Safety and Health Convention (No. 155), have long advocated for a safe working environment, which is now broadly interpreted to include psychological safety. The ILO's work, particularly in industries like maritime transport, has highlighted the unique stressors faced by seafarers, including long periods of isolation, fatigue, and the demanding nature of their work. This is highly relevant to the offshore sector, which shares many of these characteristics. Similarly, the International Maritime Organization (IMO), in its Maritime Labour Convention (MLC), has established standards for seafarer health protection, medical care, and social security. While not exclusively focused on mental health, these standards provide a foundation for national regulations and company policies to ensure the overall well-being of crew members, including their psychological state.

The IMO's work on fatigue management and crew welfare directly informs the necessity of programs like Pre-Flight Check-In. The confluence of these international frameworks—from ISO's management system approach to the ILO and IMO's focus on seafarer-specific conditions—creates a compelling case for the systematic, proactive management of mental health. These normative milestones prepare companies for a paradigm shift, moving beyond the concept that a worker is a static element of a system and recognizing that their state, in its entirety, directly interferes with operations, potentially being the difference between a successful operation and a catastrophic error. These international frameworks provide the theoretical and legal foundation for initiatives that integrate mental health into a company's core safety and risk management strategies, compelling companies to measure and mitigate the risks and damages related to the psychosocial dimension of their workforce.

The Pre-Flight Check-In program emerges as a response to this scenario, integrating elements of Psychology and Psychosocial Risk Management to help prevent the worsening of workers' mental health and assist in operational safety with regard to the individual factor. By connecting the subjective dimension of the worker's experience with objective risk indicators, the methodology enables the creation of an integrated system for individual, collective, and organizational attention. Among the main risk factors identified in the literature and in professional offshore practice are: physical and emotional isolation, sleep deprivation, physical and mental exhaustion, lack of a support network during onboard shifts, low perception of psychological safety, among others. Each of these elements can act in isolation or synergistically in the production of emotional distress. Based on this context, the Pre-Flight Check-In proposes an approach based on three pillars:

Step 01: Mandatory electronic screening; an emotional triage and risk classification stage.
Step 02: Active contact by a psychologist, in case of perceived vulnerability, for support and a better understanding of symptoms and contexts.
Step 03: Critical vulnerability cases generate a notification to the Safety Management System, for integrated decision-making with the management and health departments.

This model represents an update in how the offshore sector understands, monitors, and responds to the emotional vulnerabilities of its professionals.

Most Frequent Issues

When analyzing the main complaints reported by workers in the Pre-Flight Check-In program, the most notable complaint at present is anxious symptoms, which represent 30.6% of the moderate and alert cases recorded, surpassing other recurring complaints such as family issues (19%) and stress symptoms (13.4%).

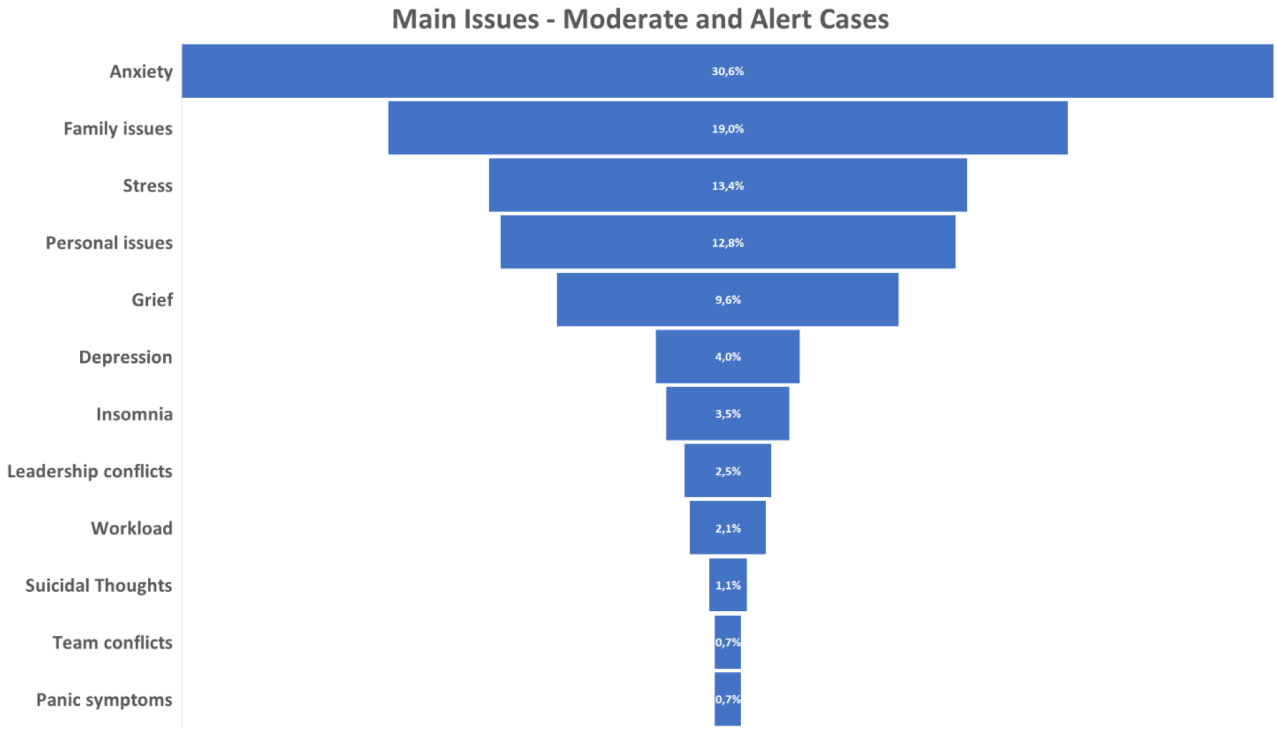

Figure 01—Source: System 4 Nuvem, CEPEM care management system, 2025)

Among the most critical symptoms perceived during the four years of the program, suicidal ideation (although not quantitatively comparable to the IADC survey mentioned above) already represents more than a thousand workers with some type of suicidal thought who were normally directed to operations on the high seas, that is, without any kind of screening or support.

Another symptom frequently perceived in the offshore population is the perception/feeling of loneliness. The distress resulting from the perception of loneliness is already growing in the global population, affecting 1 in 6 individuals according to WHO data in 2025. Program data show that the rate of perceived loneliness in the offshore population increases from the second half of the embarkation period, which generates a need to consider the dimension of loneliness in any protocol for monitoring the mental health and readiness/ fatigue of the offshore employee.

Oil and Gas Industry Segments

The analysis by segment (figure 02 below) reveals important differences in the incidence of emotional symptoms and risk cases. In offshore Drilling units, the alert indices (6%) are higher than the general average, indicating psychosocial contexts that require increased attention and suggesting that the type of operation influences the individual's emotional state.

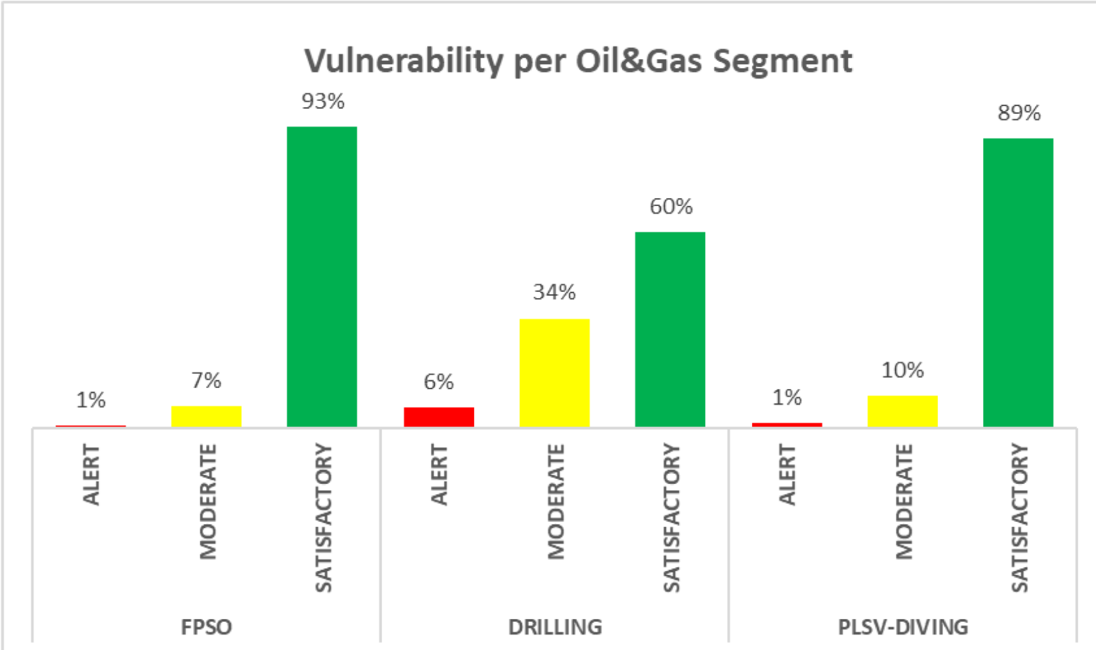

Figure 02—Source: System 4 Nuvem, CEPEM care management system, 2025)

In DRILLING, emotional distress is characterized by both the severity and the constancy and intensity of symptoms related to chronic pressure and emotional wear. The number of alert cases is 5 percentage points above the general average. Furthermore, the high percentage of stress (26%), almost on par with anxiety, indicates an environment with high operational demands, possible overload, and continuous pressure for performance—conditions that favor the escalation of symptoms to risk levels if not accompanied by adequate support measures. In addition, the Drilling segment has the highest percentage of "Moderate" cases (34%) among the three analyzed, suggesting that there is a significant contingent of workers in significant emotional distress, although not yet classified as severe, which may indicate a risk of progression if there is no early intervention. In this population, we notice 39% of workers reporting at least 03 symptoms present for at least a month, causing emotional distress. The main symptoms reported are anxiety, insomnia, and fatigue.

The PLSV-DIVING sector has the highest percentage of "suicidal thoughts" (2%) among the three analyzed segments. Diving operations are among the most demanding and dangerous in the offshore industry. They involve high levels of concentration, technical precision, and real physical risk, in addition to long periods in hyperbaric environments. This context imposes a state of constant vigilance, which can result in chronic mental fatigue, emotional exhaustion, and a sense of vulnerability, all of which are psychosocial factors that can increase occupational stress and impact emotional health. Furthermore, PLSV-DIVING has the highest percentage of anxiety (38%) compared to the other segments. Anxiety is the most prevalent symptom in the three analyzed segments, reaching 28% of cases in DRILLING and FPSO, and 38% in PLSV-DIVING. Stress is the second most incident symptom in DRILLING (26%), while it appears less frequently in FPSO (13%) and PLSV-DIVING (13%).

The increase in this symptom in DRILLING may be related to operational pressure and the demand for technical performance under complex and unpredictable conditions. Symptoms indicative of more severe psychological distress—such as depressive symptoms, suicidal thoughts, and panic attacks—are present in all three analyzed segments, although with different intensities. FPSO units show 4% depressive symptoms, 1% suicidal thoughts, and 1% panic, while the PLSV-DIVING sector records 3% depressive symptoms, 2% suicidal thoughts, and 1% panic.

These percentages not only indicate the existence of intense emotional distress but also reinforce the urgent need for institutional strategies aimed at prevention, support, and early intervention. Symptoms classified as "Family Issues" and "Personal Issues" appear with significant frequency, totaling 31% in DRILLING, 31% in FPSO, and 33% in PLSV-DIVING. This data corroborates the hypothesis that workers' emotional distress is not restricted to work dynamics but incorporates elements from their private lives, which indicates a broader field of emotional vulnerability, which is also considered and identified in the emotional check-in methodology.

Fatigue and Tiredness

The relevance of the topic of tiredness and fatigue among offshore workers, especially after shift changes or during periods of high operational demand, led to the development of a structured fatigue monitoring protocol, integrated into the emotional check-in. Unlike approaches that are limited to measuring physiological indicators, this protocol offers an active care response to the worker, also considering psychosocial and contextual aspects. Loneliness is a factor that influences both the perception and the recovery from stress, and is therefore an important factor in fatigue management. According to Soares et al. (2013), establishing a support network is essential to open a channel of connection through which the worker can find support for managing states of stress and fatigue. The methodology adopted consists of three main stages:

1. **Electronic Monitoring**: Electronic assessment of fatigue, sleep, mood, and stress in the second half of the work period.
2. **Proactive psychological intervention**: Moderate to severe cases trigger psychological support to investigate factors such as the support network and family context.
3. **Organizational Alerts**: Critical cases generate immediate notification to health for integrated decision-making with management.

Based on more than one hundred and fifty thousand (150.000) check-ins performed up to July 2025, the methodology allowed the identification that 9.5% of workers presented significant levels of fatigue during the onboard period. By integrating psychosocial data with a structured screening and care process, the protocol contributes to the prevention of undesirable events and the strengthening of a psychological safety culture in the offshore environment.

Onshore X Offshore Context

A significant finding of the program was the perception that emotional vulnerability in the administrative sector (Onshore) may be higher than that of offshore workers. The data indicate that, although operational units (Offshore) present more cases at an alert level, Onshore records a higher number of moderate cases.

This reality challenges the myth that administrative work is inherently less stressful and highlights the urgency of strengthening emotional health support for these administrative base/onshore support professionals. In a sample of 2.173 administrative/onshore workers who received check-ins, 13% presented some degree of emotional vulnerability, with 12% classified as moderate cases and 1% as alerts. The number of administrative workers with moderate vulnerability is 5 percentage points above that of offshore workers.

This data indicates that the administrative environment of offshore companies also exerts high pressure on employees due to the dynamism and complexity of the operations assisted by support functions. We can infer that the dynamism and availability required to keep up with the complexity of offshore operations generates observable wear and tear on workers at the Administrative Bases, which is above the general average of psychic vulnerability of the company itself.

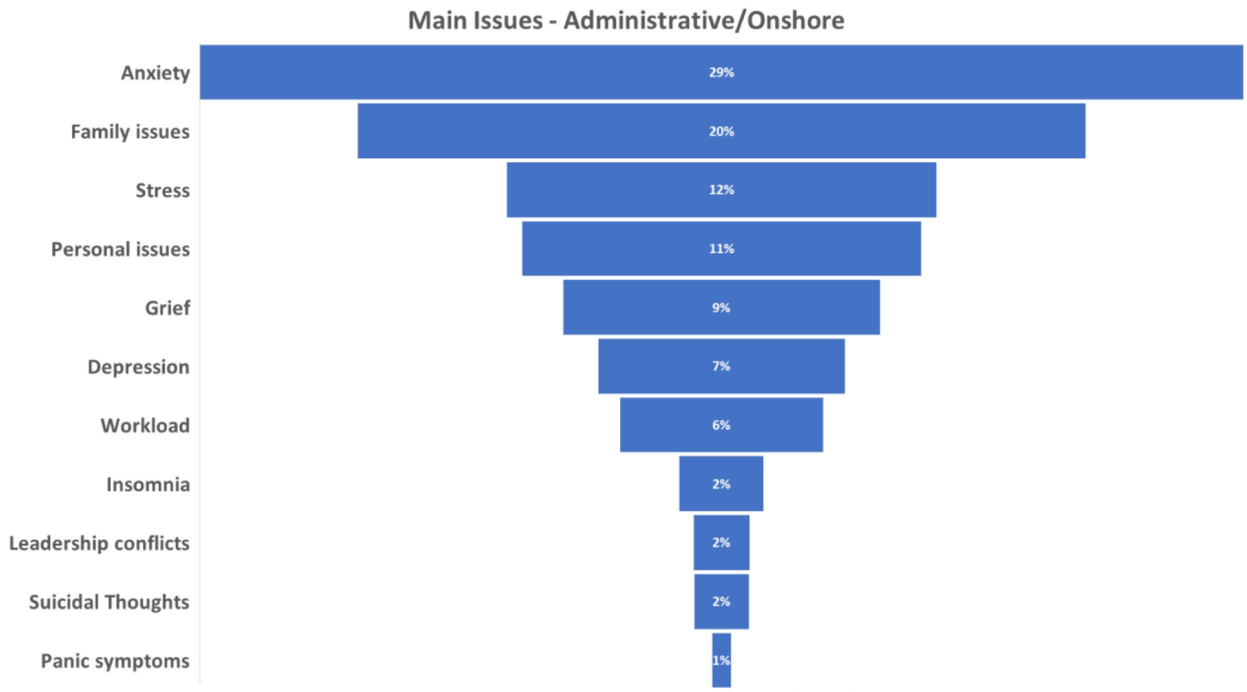

Figura 03—Source: System 4 Nuvem, CEPEM care management system, 2025)

Results

The analysis of the Pre-Flight Check-In program shows significant advances in how mental health has been integrated into operational risk management in the offshore sector. The data indicate concrete impacts at various levels of the organization, which can be summarized in the following points:

- Progressive annual increase in companies' psychological safety indicators;
- Creation of emotional risk maps by function, unit, and shift;
- A 10-fold increase in spontaneous searches for support channels;
- 105 embarkations prevented after the confirmation of critical cases;
- A 13% reduction in the perceived loneliness rate during embarkation;
- 99.7% positive recommendation of the program from employees who emphasize the latent need for direct support for offshore workers and their families.

Conclusion

The Brazilian experience with the Pre-Flight Check-In program demonstrates a transformative approach to mental health in the offshore industry, a sector historically marked by silence and a lack of organizational visibility. This initiative is aligned with a global and progressive trend of recognition and regulation of psychosocial risk factors at work. International frameworks such as ISO 45001 (2018), which included the active participation of workers and attention to mental health, and ISO 45003 (2021), which delved exclusively into psychological health at work, provided the guidelines for this new perspective. Furthermore, the official recognition of burnout by the World Health Organization (WHO) in 2022 as a work-related condition underscores the urgency of proactive interventions.

By anticipating crises and adopting an active stance of listening and support, the program acts as a collective protection device and a means of strengthening safety culture. The importance of this issue is highlighted by WHO data, which indicates that global mental health intensified by 25% after the COVID-19 pandemic. The early detection of an emotionally vulnerable worker not only prevents incidents and saves lives but also signals a paradigm shift in labor relations and the organization's role, which begins to consider the operator's state as a whole as a factor that directly contributes to system safety.

The value of identifying when a person becomes a risk to themselves, to the operation, and to the collective—and, even more, of acting before that risk materializes—is immeasurable. The management of psychosocial risks, although dealing with an "invisible risk," demands a new organizational competence: the ability to care for the invisible with responsibility and method. This requires the work system to be capable of absorbing vulnerabilities and demonstrating proactivity through coordinated actions to educate and mitigate crises.

The Brazilian model has already demonstrated technical maturity and feasibility for expansion to other high-complexity operations globally, such as mining, renewable energy, and aviation. The positive results stem from the combination of technology, empathetic listening, and data-driven strategic management, reaffirming that investing in mental health is increasingly indispensable in our operations. The future lies in the intelligent combination of technology and human contact, establishing consistent channels for expressing vulnerability to transform the topic into a strategic axis of management.

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